

# CS 411/507 - Cryptography\*

## Fall 2025 - 2026

### Instructor Information

*Name:* Erkay Savaş  
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*Office Hours:* Wednesday 16:40-18:30 (FENS 2037)

### Schedule

*Lectures:* Wednesday 18:40 - 19:30 FMAN G071  
Thursday 17:40 - 19:30 FMAN G071

This is a three-credit introductory course on the methods, algorithms, techniques, and tools of data security and cryptography. After studying the theoretical aspects of cryptographic algorithms and protocols, we show how these techniques can be integrated to solve particular data and communication security problems. This course material is of use to computer and communication engineers who are interested in embedding security into an information system, and thus, providing integrity, confidentiality, and authenticity of the documents and the communicating parties.

### Catalogue Data

This is an introductory course on cryptography. Topics include: Classical cryptosystems, basics of number theory, symmetric key cryptography (stream and block ciphers), hash functions, public key cryptosystems (RSA, discrete logarithm based algorithms, and elliptic curve cryptography (ECC)), digital signatures, implementation issues, secure key establishment techniques, secret sharing, and zero-knowledge proof.

### Prerequisite

CS 411 is open to any undergraduate students, who have previously taken MATH 204 – Discrete Mathematics - and scored minimum grade of D. Additionally, some experience in Python programming language is required.

CS 507 is open to graduate students. There is no prerequisite for CS 507.

### Tentative Outline

- **Introduction and Classical Cryptosystems:** Secure communication. Attacks to cryptosystems. Classical cryptographic techniques and algorithms. One-time pad, randomness and pseudo-randomness
- **Mathematical Foundations:** Number theory. Finite fields. Primitive roots. Exponentiation and discrete logarithm.

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\*All information in this document is tentative. The instructors reserve the right to make changes in the semester.

- **Secret-Key Cryptography:** Block ciphers and stream ciphers. DES, AES (Rijndael). Modes of operation.
- **Public-Key Cryptography:** One-way functions. Trapdoor one-way functions. Public-key cryptosystems. RSA, Diffie-Hellman, ElGamal, and elliptic curve cryptosystems.
- **Authentication and Digital Signatures:** Hash functions and message-digest functions. Digital signatures. Authentication protocols, forward secrecy, plausible deniability.
- **Protocols:** Secret Sharing Schemes, Zero-knowledge proof systems. Key management architectures, Public Key Infrastructure.

### Textbook

Nigel P. Smart. Cryptography Made Simple. Springer, 2016. ISBN 978-3-319-21936-3

### References

- W. Trappe and Lawrence C. Washington, Introduction to Cryptography with Coding Theory. 2<sup>nd</sup> Edition, Prentice Hall, 2006.
- C. Paar and J. Pelzl, Understanding Cryptography, Springer 2010
- A. J. Menezes P. C. van Oorschot, and S. A Vanstone. Handbook of Applied Cryptography, CRC Press, 1997.
- D. R. Stinson, Cryptography: Theory and Practice, 3<sup>rd</sup> Edition, Chapman Hall/CRC, 2006.

### Student Responsibilities

- **Exams:** Students are required to comply with instructor's rules for exams (quizzes, midterm, and final)
- **Homework assignments:** There will be about four homework assignments. You will be required to write programs in Python programming language.
- **Term project:** Students are required to work on a term project. It is essential for students to meet time schedule of the projects. Project groups must provide a demonstration/presentation of their work. During the demonstration/presentation, all the project members must be present. Students may work in groups of two.

### Tentative Grading

- Midterm Exam 30%
- Final Exam 40 %
- Term Project 15%
- Homeworks (Total 4( $\pm 1$ )) 15%

The letter grade boundaries will be determined by the instructors at the end of the term. In order to pass the course, the overall grade has to be higher than D/F boundry and the final exam grade has to be higher than 25/100.

Exams will be held in-person.

## Exam Dates

*Midterm:* TBA (possibly 8<sup>th</sup> or 9<sup>th</sup> week of the term)

*Final:* will be scheduled by SR

## Make-up Policy

- There will be no make-up for homeworks, and term project. Students automatically get 0 (zero) from the respective assignment grade if any of them is missed.
- Make-up is only allowed for the midterm and final examinations to those with an official medical report and to those with an official permission notice from the university on the date of the exam in question.
- Make-up examinations may be written and/or oral.

## Time Conflict Permission Policy

In general, time conflict permission requests are approved in this course <sup>1</sup>. However, time conflict permission is not a permission for you not to attend some classes. Moreover, it does not mean that, time conflicts will be taken into account when planning any activity (exam, quiz, etc.) for the corresponding hour. It is the responsibility of the student, who took time conflict permission, to manage potential problems that may arise due to time conflict.

It is assumed that the students, who are registered to this course with time conflict permission, accept these terms.

## Plagiarism Policy (Academic Integrity)

Plagiarism means presenting someone else's work as yours. This is a very serious and ethical problem. A plagiarized work may or may not be a verbatim copy of another submission. Verbatim copies are of course plagiarized ones. However, if a submission is derived from another one by partially changing or rephrasing some parts, this action is also plagiarism. When a plagiarism case is detected, sanctions are applied to all parties regardless of the actual source of the submission. These sanctions are as follows:

- For the midterm/final examinations,
  - students directly fail the course, even in the first offense.<sup>2</sup>
- For the homeworks and term project,
  - for the first time, all plagiarized submission owners receive 0,
  - the second time, the student fails the course automatically.<sup>3</sup>

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<sup>1</sup>The instructors reserve the right to reject any request.

<sup>2</sup> Additionally, the case will be referred to the FENS Dean's Office for disciplinary action. This course does not tolerate any breach of academic integrity (more info on <https://www.sabanciuniv.edu/en/academic-integrity-statement>)

<sup>3</sup>See footnote 2